

Daniel Cohen

Mathematics & Statistics Professional | Quantitative Analytics | Finance | Data & AI Applications
GitHub: github.com/Coakley11

PROFESSIONAL SUMMARY

Mathematics and Statistics professional with an M.A. in Statistics & Applied Mathematics (GPA 3.93), MBA in Finance & Investments (Finance GPA 4.00), actuarial exam foundation (SOA P, FM, MFE), and a deployed analytics portfolio spanning Python dashboards, SQL, Excel, and AI evaluation workflows. Translates probability, hypothesis testing, investment analysis, and quantitative reasoning into decision-support applications - not just classroom theory.

EDUCATION

M.A. Statistics & Applied Mathematics - Hunter College

GPA: 3.93

Overall GPA: 3.88 | Finance GPA: 4.00

Magna Cum Laude | Arthur D. Gayer Memorial Award in Economics

CREDENTIALS AND CERTIFICATIONS

- Society of Actuaries - Exam P/1 (Probability), FM/2 (Financial Mathematics), MFE (Models for Financial Economics)
- NYS Certification - CST Mathematics (004), Educating All Students (EAS)

EXPERIENCE

Analytics & AI Portfolio Developer (2024 - Present)

Independent - Daniel AI Suite

- Built and deployed seven production Streamlit applications for portfolio analytics, sports modeling, AI evaluation, simulation, and decision-support workflows
- Authored SQL & Excel analytics workbooks with KPI dashboards, pivot exercises, and 42+ structured SQL queries including AI evaluator datasets
- Designed case-study narratives and demo modes translating statistical methods into recruiter-ready business interpretation

Full-Time High School Mathematics Teacher (2024 - Present)

The Montfort Academy

- Instruct Algebra II, Geometry, and Pre-Calculus with emphasis on quantitative reasoning, functions, and data interpretation
- Support Chemistry and Physics laboratory sessions requiring measurement, precision, and analytical documentation
- Apply structured problem-solving frameworks that parallel analytics debugging and hypothesis-driven analysis

Permanent Full-Time Substitute Teacher (2021 - 2024)

Yeshiva Har Torah

- Delivered K-8 secular and Judaic studies instruction across multiple subjects and grade levels
- Prepared science laboratory materials and supported hands-on quantitative observation activities
- Maintained classroom data tracking for student progress and instructional adjustment

Mathematics Instructor (2020)

Pace University

- Taught Mathematics for Life, College Algebra, and Calculus I - connecting algebraic reasoning to applied decision contexts
- Emphasized interpretation of quantitative results, not just procedural computation

Mathematics Instructor (2020)

Yeshiva University

MBA Finan

B.A. Math

- Led Calculus III Problem Solving Seminar focused on multi-step analytical reasoning and proof-style logic

Full-Time Lecturer / Adjunct Lecturer (2014 - 2019)

Kingsborough Community College (CUNY)

- Taught Quantitative Reasoning, College Algebra, Intermediate Algebra, Principles of Mathematics, and Pre-Algebra
- Developed curriculum reinforcing probability intuition, algebraic modeling, and statistical literacy at scale
- Supported diverse learners in translating word problems into structured quantitative frameworks

Adjunct Lecturer (2015 - 2019)

Queens College (CUNY)

- Instructed Probability & Statistics, Pre-Calculus, Calculus, College Algebra, and Math for Elementary School Teachers
- Taught hypothesis-testing concepts, probability distributions, and regression intuition at undergraduate level

Business Statistics Lab Professor (2012)

Queens College

- Led Business Statistics labs with Excel instruction, computer-lab analytics exercises, and data interpretation assignments
- Connected descriptive statistics, inference, and spreadsheet modeling to business decision scenarios

College Assistant & Academic Tutor (2011 - 2014)

Queens College

- Tutored Statistics, Economics, and Finance with focus on quantitative support and exam preparation
- Diagnosed knowledge gaps and built targeted practice for data interpretation and financial quantitative reasoning

Mathematics & Statistics Tutor (2011 - Present (periodic))

Tutor Doctor | Club Z | Private | SAT Prep

- Provided individualized instruction in Mathematics, Economics, Statistics, and Finance from remedial through advanced undergraduate levels
- Delivered SAT mathematics preparation emphasizing problem decomposition and analytical speed

SELECTED PROJECTS

Investment Portfolio Analyzer | Python | Streamlit | yfinance

- Built live portfolio dashboard with Sharpe/Sortino metrics, Monte Carlo simulation, efficient frontier optimization, and correlation heatmaps
- Separated analytics core from UI for maintainability; deployed on Streamlit Cloud

Baseball Analytics Explorer | Python | Streamlit | scikit-learn

- Developed MLB analytics platform with statistical significance testing, ML projections, and fantasy draft decision support
- Implemented pytest coverage for projection calibration and draft strategy logic

Applied Mathematical Intelligence | Python | Streamlit

- Designed modular Mathematical Thinking Lab with betting EV, AI training visualization, and epidemic simulation engines
- Built pedagogical UX pattern: explain -> simulate -> interpret -> math depth

SQL & Excel Analytics Workbooks | Excel | SQL | pandas

- Created AI evaluator, investment, quant, and credit-risk workbooks with KPI dashboards and SQL practice queries
- Demonstrates fluency across Python dashboards and spreadsheet-based business reporting

TECHNICAL SKILLS

Statistics | Probability | Hypothesis Testing | Python | SQL | Excel | Pivot Tables | Pandas | Streamlit | Data Visualization | Machine Learning | Monte Carlo Simulation | Portfolio Optimization | Financial Analysis | Dashboard Development | AI Evaluation | Git / GitHub | AI-Assisted Development | Business Analytics | Decision Support Systems

TARGET ROLES

Data Analyst | AI Evaluator | AI Trainer | Business Intelligence Analyst | Research Analyst | Junior Data Scientist | Quantitative Analyst | Financial Analyst